



ArchiMate Exchange Format - handle sourceAttachment and targetAttachment #1237

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Phillipus opened on Apr 29

Last edited by Phillipus ▾

Member

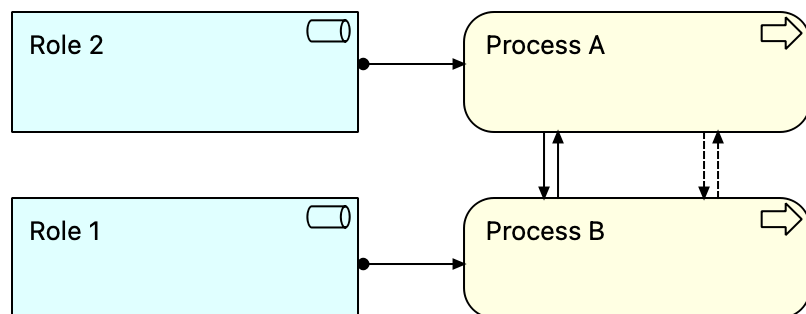


Description

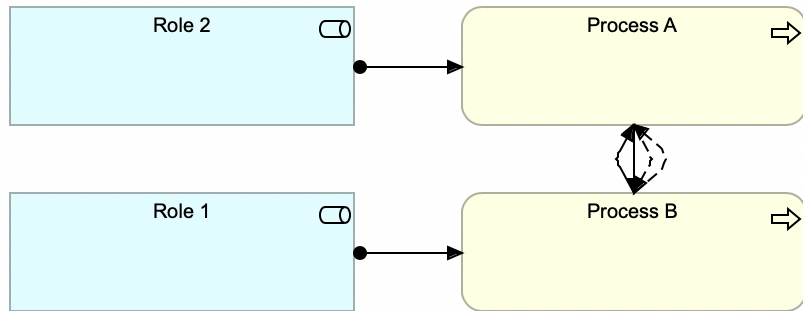
Archi doesn't support source and attachment points from the ArchiMate Exchange Format on connections. It might be possible that when importing an exchange file to add bend points to create a similar effect.

Example

Exported from:



After imported:



Phillipus on Apr 29

Member

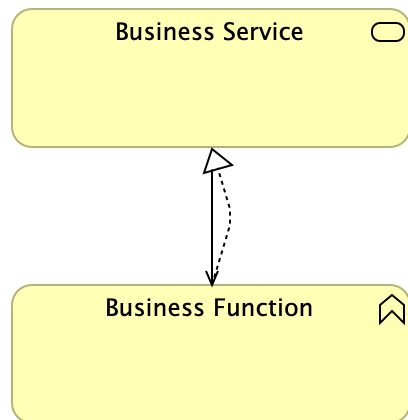
Author



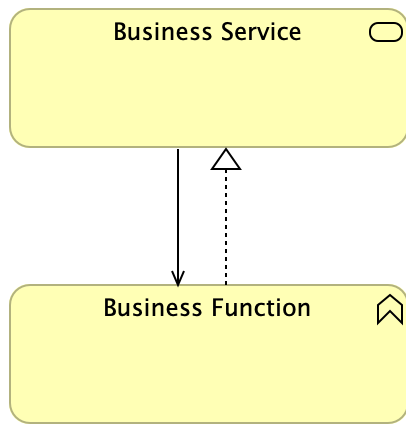
Here's a simple Exchange format file:

[bendpoint test.xml](#)

It looks like this when imported:



Aim to make it look like this:



[bendpoint test.archimate.zip](#)



Phillipus on Apr 29

Member

Author



If anyone would like to contribute a PR please look [here](#) in the code.



Phillipus added [help wanted](#) on Apr 29



jbsarrodie on Apr 29

Member



Hi, as I'd like to work on supporting some of the features of the upcoming new version of the exchange file format (which will be needed at some point to support ArchiMate 4), I might have a look at that too. No promise though, so anyone willing to help is welcome.

I'm going to reinstall all my dev stack soon, maybe with a bit of AI to see if that helps ;-)



mwith on May 9

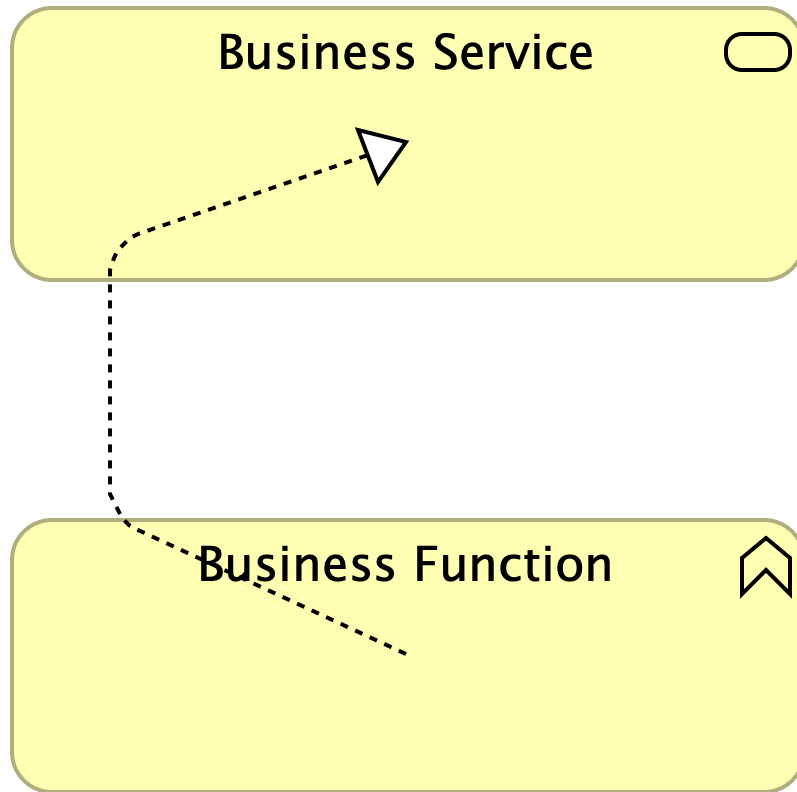
Contributor



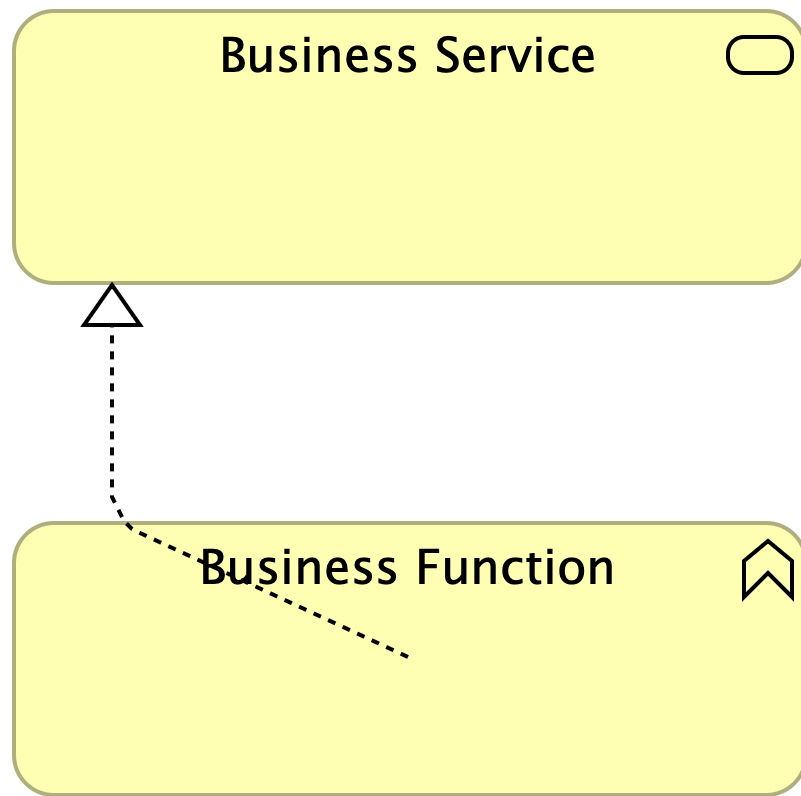
Hi Phil

I've done some experimentation implementing sourceAttachment and targetAttachment as bendpoints. I used an xml file with just one connection. This is my findings

This is with sourceAttachment and targetAttachment



With just sourceAttachment it looks like this

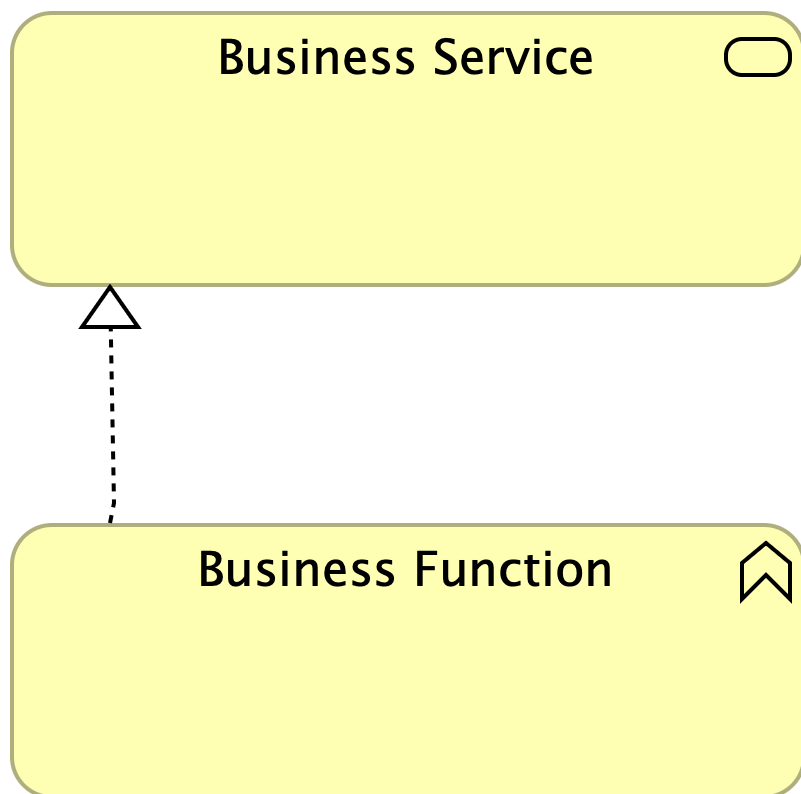


It seems that if the bendpoint is located on the edge of the object then you get the line segment from the center of the object to the bendpoint. So I tried to move the bendpoint slightly (5 on the y-axis) outside the element then I got this that looks more the wanted. But you can see the bendpoint with the curve.

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I don't see that using bendpoints is the correct solution for this. In my view the attachment points should be modelled separately on the connection and if present used by the drawing algorithm.

What is your take on this?

BTW why are bendpoints converted into lines with a start and end in the `createBendPointFromAbsolutePosition()`. How does this work for connections that have multiple bendpoints?



Phillipus on May 9

Member

Author



[@mwith](#) Thanks for looking at this. A few comments here.

In my view the attachment points should be modelled separately on the connection and if present used by the drawing algorithm.

That would require a major re-write of anchor points at the GEF level. The [OrthogonalAnchor](#) was written by [@jbsarrodie](#) and I believe it uses bendpoints to position the connection.

why are bendpoints converted into lines with a start and end in the `createBendPointFromAbsolutePosition()`. How does this work for connections that have multiple bendpoints?

I think it's because of the XML exchange format, I'd have to take a deep dive into that part of the code again.



mwith on May 9

Contributor



Point taken.

So if using bendpoints for this then I think the attachment point should be adjusted on the x- and y-axis so it is just outside the element likewise for any other bendpoints for the connection.

What is the behaviour if `sourceAttachment` or `targetAttachment` is not present in the xml file? Say `targetAttachment` is not present should it go from the anchor point on the source element towards the center for the target element or in a vertical or horizontal line depending on where the connection is attached (top, right, ...)?

I'd like to continue working on this to see if I can get the bendpoints work like expected for all variations

/Morten



Phillipus on May 9

Last edited by Phillipus ▼

Member

Author



What is the behaviour if `sourceAttachment` or `targetAttachment` is not present in the xml file?

Do nothing, just add bend-points if present.

`Draw2d` anchor points are used for connections. These are algorithm based, either `OrthogonalAnchor` if the preference is enabled or `ChopboxAnchor` if not. There are also a few sub-classes of `ChopboxAnchor` to handle odd shapes. The algorithm for each anchor point calculates the attachment points of the connection based on the source and target figures' bounds.

I'd like to continue working on this to see if I can get the bendpoints work like expected for all variations

Thanks.



Phillipus on May 9

Member

Author



One more comment.

This assumes that the Archi preference is set to use `OrthogonalAnchor` (it is on by default). If it is off then `ChopboxAnchor` and its sub-classes are used and so there is no way to simulate or set attachment points as these are calculated dynamically.



Phillipus on May 9

Member

Author



...and another comment while I'm thinking about this. ;-)

There seems to be an assumption in some quarters that ArchiMate tools must support all things in the exchange format. Many things are optional of necessity. Some tools might implement bend-points and connection attachment points and other tools might not. Some tools might support colour gradients and label positions, others not. The exchange format is an *attempt* to exchange some graphical features, *if implemented*. There is nothing in the ArchiMate spec to say that a tool has to even implement colours. So the exchange format will always be "lossy".



Phillipus mentioned this [on May 12](#)



[Import Archimate Exchange file supports sourceAttachment and targetAttachment for connections #1242](#)

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